

Sai Markapuram

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Professional Summary

- Embedded Systems and Controls Engineer with 5+ years of experience in developing, validating, and automating software for safety-critical embedded systems across automotive and robotics platforms.
- Strong background in Embedded C/C++, Python, and MATLAB/Simulink for implementing control algorithms, developing firmware interfaces, and building automated validation frameworks for real-time control systems.
- Extensive experience designing and executing Hardware-in-the-Loop (HiL) and Software-in-the-Loop (SiL) test environments using dSPACE ControlDesk, NI VeriStand, and Vector CANoe for closed-loop validation of ADAS, powertrain, and body control systems.
- Developed scalable Python-based automation frameworks for system validation, log analysis, diagnostics extraction, and test orchestration integrated with CI/CD pipelines using GitHub Actions and Jenkins to enable nightly regression testing.
- Hands-on experience developing and validating control logic for Engine Control Modules (ECM), Transmission Control Modules (TCM), and Body Control Modules (BCM) including state-machine design, fault handling, and safety monitoring logic.
- Experienced in validating ADAS features such as Lane Keep Assist (LKA), Adaptive Cruise Control (ACC), and Forward Collision Warning (FCW) through HiL simulation, scenario-based testing, and automated signal validation.
- Proficient in vehicle communication protocols including CAN, LIN, and Automotive Ethernet with deep experience using CANalyzer, CANoe, Vehicle Spy, and python-can for network diagnostics, message analysis, and fault injection.
- Designed automated regression suites that execute large-scale test scenarios, analyze telemetry logs, detect anomalies, and generate validation reports for continuous software delivery.
- Experienced in debugging embedded systems using oscilloscopes, logic analyzers, protocol analyzers, and calibration tools such as INCA and MATLAB for signal verification and real-time performance analysis.
- Collaborative engineer experienced working with cross-functional teams across systems, hardware, and software domains to debug complex issues, tune parameters, and validate control algorithms within ASPICE and ISO 26262 compliant development environments.
- Proficient in Agile development environments using Git, GitHub, JIRA, JAMA, IBM DOORS, and Confluence for requirements traceability, automated testing pipelines, and structured validation workflows.

Technical Skills

Programming & Scripting: Embedded C, Modern C++ (C++17/20), Python, Bash, CAPL, Java, MATLAB, Simulink Scripting

Automation & Validation Frameworks: Robot Framework, PyTest, unittest, Jenkins Pipelines, GitHub Actions, custom Python test harnesses for HiL/SiL automation

Communication Protocols: CAN, LIN, Automotive Ethernet, SPI, I²C, UART, UDS, OBD-II, J1939, XCP, CCP

Tools & Environments: Vector CANoe/CANalyzer, Vehicle Spy, dSPACE ControlDesk, NI VeriStand, Vector CAST, INCA, ETAS LabCar, Vector vTestStudio

Modeling & Simulation: MATLAB/Simulink, Stateflow, ROS, Gazebo, Simscape, Model Advisor, MIL/SIL/HiL closed-loop simulation setups

Powertrain & ECU Systems: BCM, ECM, TCM, ABS, Inverter, Motor Control ECUs, Diagnostics Calibration, Torque/Speed Control, CAN-based powertrain validation

CI/CD & DevOps: Git, Jenkins, GitHub Actions, Docker, Azure DevOps, Linux scripting for automated builds and test orchestration

Test Management & Traceability: Jira Xray, Zephyr, IBM DOORS, JAMA, Polarion ALM, TestLink, Requirements-to-Test mapping under ASPICE

Standards & Compliance: AUTOSAR (Classic & Adaptive), ISO 26262, ASPICE Level-2/3, MISRA-C/C++, Functional Safety & Cybersecurity (ISO/SAE 21434)

Hardware & Debug: Oscilloscope, Logic Analyzer, DMM, Protocol Analyzer, Power Analyzer, CAN loggers, fault-injection benches

Industrial & Control Systems: PLCs (Siemens TIA Portal, Allen-Bradley Studio 5000), Modbus/TCP, OPC-UA integration, SCADA data pipelines

Professional Experience

Embedded Systems Engineer

Jan 2024 – Present

General Motors – Michigan, USA

- Develop embedded firmware and control software for automotive and robotics subsystems using Embedded C/C++ on AUTOSAR Classic and RTOS-based ECU platforms.
- Design and implement control logic, diagnostics handlers, and communication stacks for ADAS and vehicle subsystems including Lane Keep Assist (LKA), Adaptive Cruise Control (ACC), Engine Control Modules (ECM), and Body Control Modules (BCM).
- Build and validate control algorithms and state-machine logic using MATLAB/Simulink with auto-code generation workflows for embedded targets.
- Perform Hardware-in-the-Loop (HiL) and Software-in-the-Loop (SiL) validation using dSPACE ControlDesk, NI VeriStand, and Vector CANoe to simulate real-world vehicle dynamics and sensor inputs.
- Develop Python-based automation frameworks to orchestrate HiL test execution, automate regression scenarios, parse ECU telemetry logs, and generate validation reports for nightly testing pipelines.
- Implement CI/CD workflows using GitHub Actions and Jenkins to enable automated firmware builds, regression testing, and system validation in continuous integration environments.
- Perform network communication validation across CAN, LIN, and Automotive Ethernet buses including signal decoding, timing verification, and fault injection using CANalyzer, Vehicle Spy, and python-can.
- Analyze ECU runtime behavior through signal traces, calibration data, and telemetry logs to debug control performance, communication issues, and real-time system anomalies.
- Integrate robotics simulation environments using Gazebo to validate perception-control interactions and test algorithm behavior under simulated sensor and environment conditions.
- Develop automated test suites for system-level validation including fault handling, diagnostics verification, and communication robustness testing across embedded controllers.
- Utilize oscilloscopes, logic analyzers, and protocol analyzers to debug firmware timing, signal integrity, and hardware–software integration issues.
- Collaborate with cross-functional teams including systems, controls, perception, and hardware engineering to validate end-to-end control software functionality and ensure compliance with ASPICE and ISO 26262 development standards.

Embedded Software Engineer

May 2022 – Dec 2023

John Deere – Waterloo, IA

- Developed embedded software for powertrain and vehicle control systems used in off-highway and heavy equipment platforms using Embedded C/C++ in real-time ECU environments.
- Implemented control algorithms, diagnostics routines, and state-machine logic for engine and drivetrain subsystems to ensure stable operation under varying load and environmental conditions.
- Designed and validated control software using MATLAB/Simulink model-based development workflows, supporting auto-code generation and real-time embedded deployment.
- Integrated embedded firmware with AUTOSAR-based and RTOS-based ECU architectures, enabling reliable communication across CAN and LIN networks.
- Performed Hardware-in-the-Loop (HiL) and bench-level validation to verify powertrain control logic, fault-handling mechanisms, and system robustness under simulated operating conditions.
- Utilized Vector CANoe, CANalyzer, and other vehicle communication tools to analyze network traffic, validate signal timing, and troubleshoot ECU communication issues.
- Developed Python-based automation scripts to execute regression test scenarios, parse ECU diagnostic logs, and generate validation reports to support firmware release cycles.
- Conducted system-level debugging using oscilloscopes, logic analyzers, and calibration tools to verify signal integrity and real-time controller responses.
- Collaborated with systems, calibration, and hardware engineering teams to tune control parameters, validate sensor interfaces, and ensure compliance with functional safety and reliability requirements.

- Maintained software modules within Git-based version control systems and supported automated build and integration pipelines for embedded firmware development.

Embedded Software Engineer

May 2019 – Dec 2020

ZF – Hyderabad, India

- Developed embedded software and firmware components for automotive electronic control units (ECUs) using Embedded C/C++ in real-time embedded environments.
- Implemented control algorithms and diagnostics logic for vehicle control systems using MATLAB/Simulink model-based development workflows with auto-code generation.
- Designed and optimized real-time software modules for vehicle control and safety functions, ensuring deterministic execution and robust fault handling.
- Integrated embedded firmware with AUTOSAR-based ECU platforms and implemented communication interfaces across CAN, LIN, and Automotive Ethernet networks.
- Developed low-level software components for sensor interfacing, actuator control, and signal processing within automotive control systems.
- Performed Software-in-the-Loop (SiL) and Hardware-in-the-Loop (HiL) validation to verify control logic, communication timing, and system stability.
- Utilized Vector CANoe and CANalyzer to debug vehicle network communication, analyze message timing, and validate ECU signal interactions.
- Developed Python-based utilities to automate validation workflows, process ECU diagnostic logs, and support regression testing for firmware releases.
- Debugged embedded software issues using oscilloscopes, logic analyzers, and communication trace tools to diagnose timing issues and signal integrity problems.
- Collaborated with systems, hardware, and calibration teams to integrate software modules, tune control parameters, and validate system behavior under different operating conditions.
- Maintained embedded software modules within Git-based repositories and supported build and integration processes for ECU firmware development.

Education

Master of Science in Information Technology, Cleveland State University – Cleveland, Ohio, USA

Bachelor of Technology in Electronics and Communication Engineering, Jawaharlal Nehru Technological University Hyderabad (JNTUH) – Hyderabad, India

Academic Projects

Cloud-Based Inventory Management System

- Developed a full-stack inventory management application using Python, Flask, and MySQL for tracking products, orders, and stock levels.
- Implemented REST APIs and role-based authentication to manage access for administrators and users.
- Deployed the application on AWS EC2 with database integration and automated backup strategies.

Data Analytics Dashboard for Business Insights

- Built an interactive data analytics dashboard using Python, Pandas, and Power BI to analyze sales and operational data.
- Designed ETL pipelines to clean, transform, and visualize large datasets for business decision-making.
- Generated reports and KPIs including revenue trends, product performance, and customer segmentation.